

FSOC Coarse Pointing, Acquisition & Tracking - Performance Report

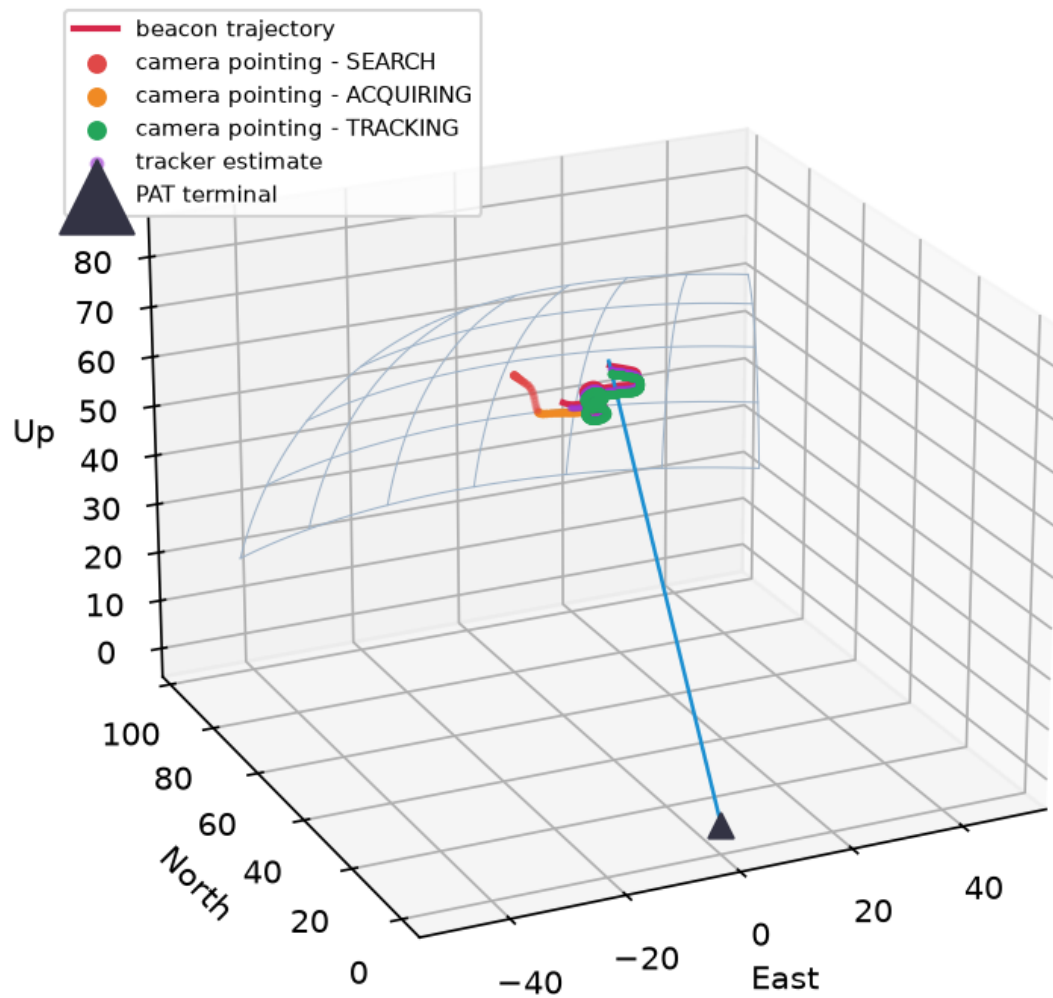
Run	
Generated	2026-10-02T16:32:22
Seed	8
Identification method	hybrid
Beacon path	random_walk at 1.51 deg/s
Decoys	5 (4 blinking)
Turbulence / vibration	0.6 / 0.43
Clouds	yes
Reported-position error (GPS prior)	3.58 deg
Camera start	az 0.0, el 0.0 deg (9.5 deg from the reported position)

Performance summary

Metric	Value
Simulation duration	20.0 s
Frames simulated	600
Achieved frame rate	35.6 FPS
Processing time (mean / max)	6.44 / 14.81 ms
Processing throughput	155 FPS
Acquisition time	2.07 s
Time locked on beacon (whole run)	89.7 %
Lock retention (after first lock)	100.0 %
Lock retention while beacon visible	100.0 %
Lock losses	0
Tracking error mean / RMS / max	4.66 / 5.45 / 12.52 px
Tracking error mean / max	1.017 / 2.732 mrad
Ready for fine-pointing handover	98.0 % of locked time
False-lock frames	0
Candidates rejected / locks dropped	0 / 0
Beacon in field of view / occluded	97.3 / 52.3 %

3D view of the run

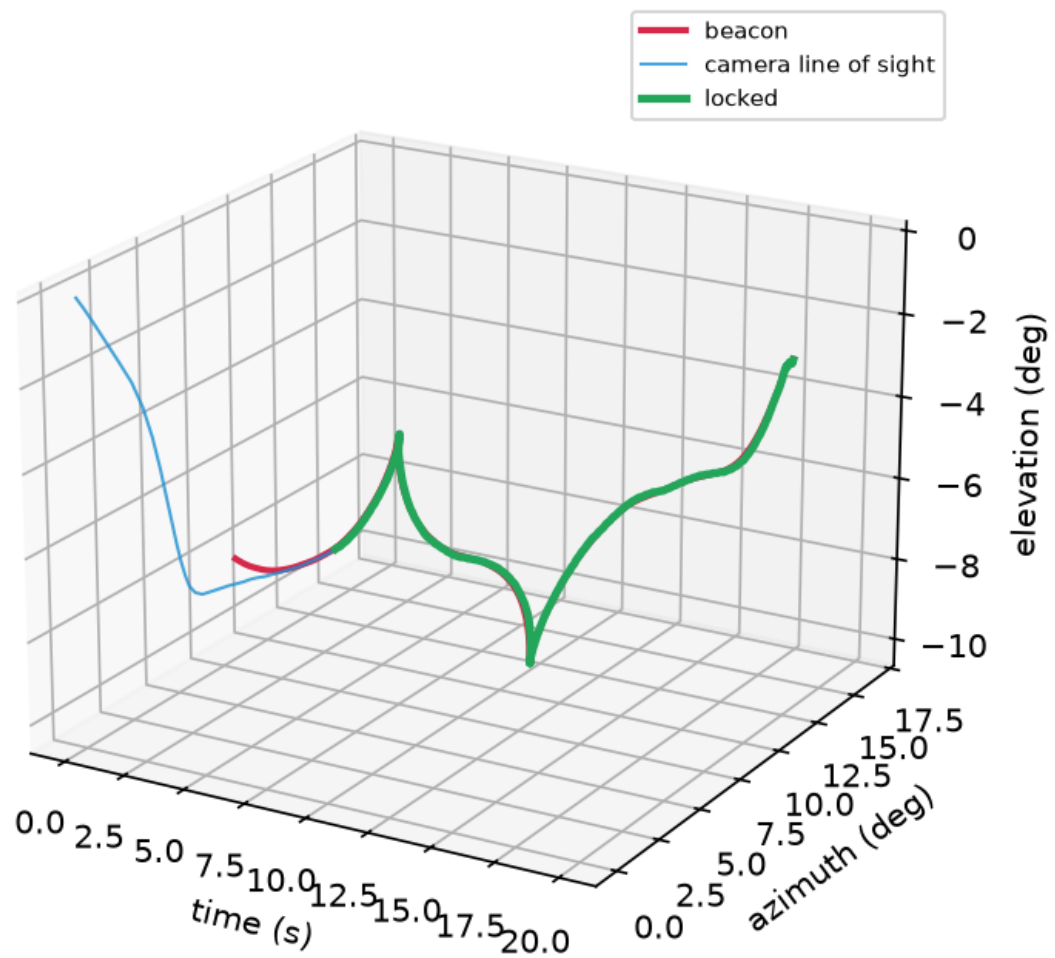
3D view: beacon trajectory and camera line of sight on the sky dome



Red: true beacon path. Dots: where the camera pointed, coloured by tracker state (red search, orange acquiring, green tracking, yellow coasting). Purple: tracker estimate.

3D view: azimuth / elevation over time

3D view: azimuth / elevation over time - beacon vs camera pointing



The camera line of sight (blue, green while locked) converging onto and following the beacon (red) shows acquisition and tracking in one picture.

Time charts

Tracking performance over time

